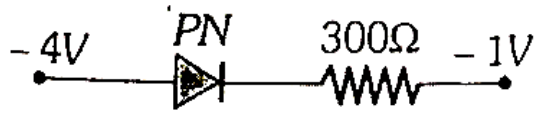
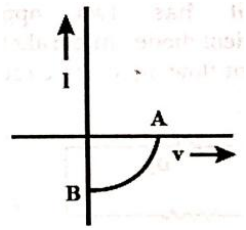
Semiconductor

1. In a semiconductor, the concentration of electrons is $8\times10^{14}/\text{cm}^3$ and that of the holes is $5\times10^{12}/\text{cm}^3$. The semiconductor is
   1) P-type
   2) N-type
   3) Intrinsic
   4) PNP-type
2. Although carbon, silicon and germanium have same lattice structure and four valence electrons each, their band structure leads to the energy gaps as
   1) $E_g(\text{Si}) < E_g(\text{Ge}) < E_g(\text{C})$
   2) $E_g(\text{Si}) > E_g(\text{Ge}) < E_g(\text{C})$
   3) $E_g(\text{Si}) < E_g(\text{Ge}) > E_g(\text{C})$
   4) $E_g(\text{Si}) > E_g(\text{Ge}) > E_g(\text{C})$
3. Which of the following statements is true for an N-typed semi-conductor
   1) The donor level lies closely below the bottom of the conduction band
   2) The donor level lies closely above the top of the valence band
   3) The donor level lies at the halfway mark of the forbidden energy gap
   4) None of above
4. A p-n photodiode is fabricated from a semiconductor with a band gap of 2.5eV. It can detect a signal of wavelength
   1) $6000\,\text{Å}$
   2) $4000\,\text{nm}$
   3) $6000\,\text{nm}$
   4) $4000\,\text{Å}$
5. What is the current in the circuit shown below



   1) 0 A
   2) 10^{-2} A
   3) 1 A
   4) 0.10 A
6. The given graph represents *V-I* characteristic for a semiconductor device



   Which of the following statement is correct

1) It is for a photodiode and points A and B represent open circuit voltage and current, respectively
2) It is for a LED and points A and B represent open circuit voltage and short circuit current respectively.
3) It is V - I characteristic for solar cell where point A represents open circuit voltage and point B short circuit current
4) It is for a solar cell and points A and B represent open circuit voltage and current, respectively

7. The barrier potential of a p-n junction depends on

a. Typed of semiconductor material
b. Amount of doping
c. Temperature

Which one of the following is correct

1) b and c only
2) a, b and c
3) a and b only
4) b only

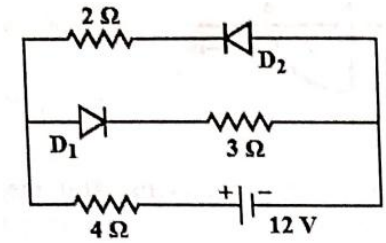
8. Depletion layer at a p-n junction contains

1) Mobile positive charges only
2) Mobile negative charges only
3) Immobile negative charges only
4) Both positive and negative immobile charges

9. A Zener diode has break down voltage of 5.0 V. The resistance required to allow a current of 100 mA through the Zener in reverse bias when connected to a battery of emf 12 V is

1) 50Ω
2) 70Ω
3) 100Ω
4) 150Ω

10. The circuit has two oppositely connected ideal diodes in parallel. What is the current flowing in the circuit



1) 2.0 A
2) 1.33 A
3) 1.71 A
4) 2.31 A

11. The logic gates output '1' for the inputs of '1' and 'O' are

1) AND and OR
2) OR and NOR

3) NAND and NOR
4) NAND and OR

12. The following truth table corresponds to the logic gate

A	0	0	1	1
B	0	1	0	1
X	0	1	1	1

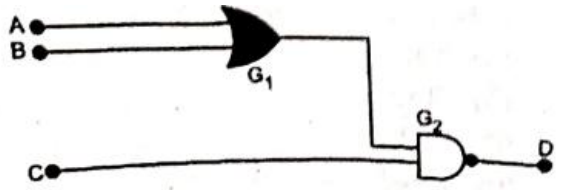
1) NAND
2) OR
3) AND
4) XOR

13. A truth table is given below. Which of the following has this type of truth table

A	0	1	0	1
B	0	0	1	1
Y	1	0	0	0

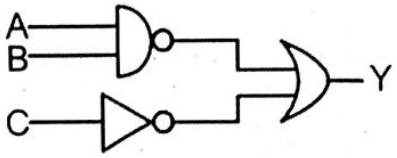
1) XOR gate
2) NOR gate
3) AND gate
4) OR gate

14. For the given combination of gates, if the logic states of inputs A, B, C are follows $A = B = C = 0$ and $A = B = 1, C = 0$ then the logic states of output D are



1) 0,0
2) 0,1
3) 1,0
4) 1,1

15. The inputs to the digital circuit are shown below. The output Y is.



1) $A + B + \overline{C}$
2) $(A + B)\overline{C}$
3) $\overline{A} + \overline{B} + \overline{C}$
4) $\overline{A} + \overline{B} + C$

16. If A and B are two inputs in AND gate, then AND gate has an output of 1 when the values of A and B are

1) $A = 0,\ B = 0$
2) $A = 1,\ B = 1$
3) $A = 1,\ B = 0$

4) $A = 0, B = 1$

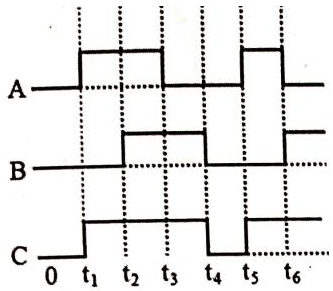
17. Which logic gate produces LOW' output when any of the inputs is HIGH'

1) AND
2) OR
3) NAND
4) NOR

18. Which of the following logic gate is an universal gate

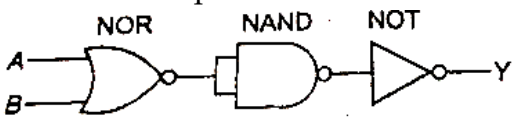
1) OR
2) NOT
3) AND
4) NOR

19. The figure shows a logic circuit with two inputs A and B and the output C. The voltage wave forms across A, B and C are as given. The logic circuit gate is



1) OR gate
2) NOR gate
3) AND gate
4) NAND gate

20. The circuit is equivalent to



1) NOR gate
2) OR gate
3) AND gate
4) NAND gate

Answer Keys

(1)	2	**(16)**	2
(2)	2	**(17)**	4
(3)	1	**(18)**	4
(4)	4	**(19)**	1
(5)	1	**(20)**	1
(6)	3		
(7)	2		
(8)	4		
(9)	2		
(10)	3		
(11)	4		
(12)	2		
(13)	2		
(14)	4		
(15)	3		